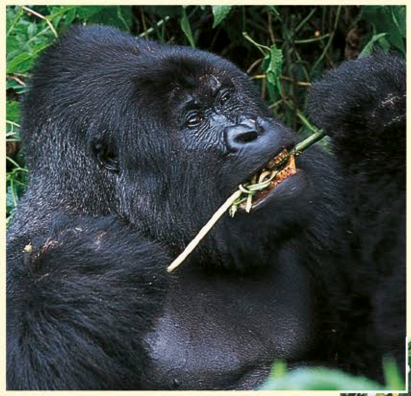
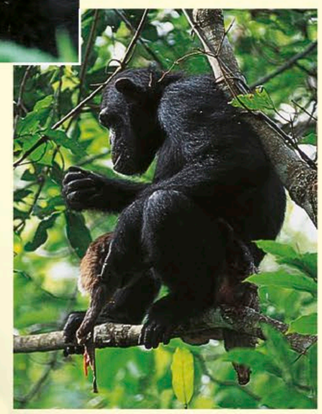
Feeding

As a general rule, small primates tend to eat insects, whereas larger species mostly eat leaves and fruit (a large primate cannot sustain itself on insects alone). Small primates have a high metabolic rate and cannot afford the long digestion times needed to process vegetable matter. Some leaf-eating species, such as colobus monkeys and langurs, have a complex stomach containing bacteria to ferment cellulose; other primate species have bacteria in the cecum or in the colon. A few species, including chimpanzees and baboons, hunt vertebrate prey as well as eating vegetable matter. Only tarsiers are entirely carnivorous.



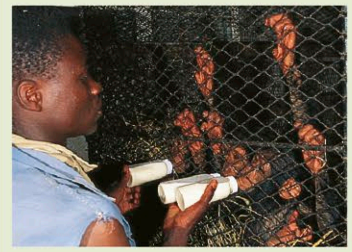
EATING VEGETATION
Of all primates, gorillas probably eat the most plant material. In order to break down cellulose (thereby releasing the cell nutrients), gorillas have large molar teeth and strong jaw muscles for chewing. Their pot belly houses a long digestive tract.



EATING MEAT
Given the opportunity, some primates will hunt (sometimes cooperatively) and kill other animals. This common chimpanzee is eating part of a duiker.

CONSERVATION

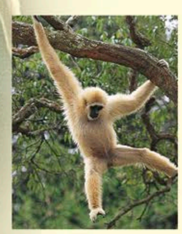
Primate numbers are declining severely in the wild due to habitat loss and, more recently, due to the bushmeat trade that includes the illegal hunting of protected species (such as the gorilla) for meat. As a result, many species are now endangered. Primates were are also widely used in medical and space research. Although a few species are being reintroduced to the wild from zoos with captive breeding programs, such as this chimpanzee orphanage in Zambia (below), the situation remains bleak.



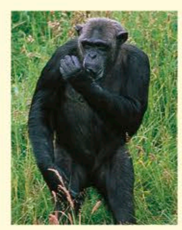
LIFE IN THE TREES
These douc langurs are typical primates in that they are arboreal, have prehensile hands and feet, live in groups, and eat leaves and fruit. Their colorful markings help provide camouflage. Like almost all primates, loss of habitat is a major threat to their survival.

Movement

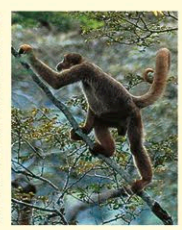
Most primates spend at least part of their life in trees and have adapted accordingly. To provide a strong hold on branches, the big toe is separated from the other toes in all species except humans, and the thumb is always separated from the fingers, although it is fully opposable (that is, it can turn, face, and touch the other digits in the same hand) only in apes and in some Old World monkeys. The arm and wrist bones are not fused, which increases dexterity. Primates also have "free" limbs—the upper part of each limb is outside the body wall, which allows great freedom of movement. (In other mammals, such as horses, the upper part is inside the body wall—the "armpit" is in fact the elbow joint, which makes movement more restricted.) Some species have a long, prehensile tail, used as a "fifth limb."



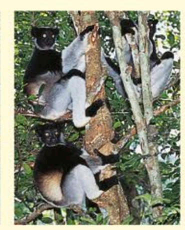
SWINGING
Spider monkeys and gibbons, such as this lar gibbon, use their long arms to swing from branch to branch. This is called brachiation.



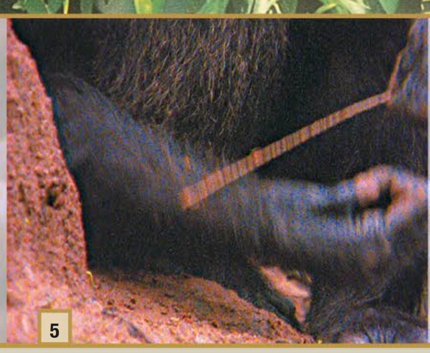
STANDING
Primates, such as chimpanzees, that are capable of standing and walking on 2 legs (known as bipedalism), tend to have long legs.



CLIMBING
The most common way of moving is on all fours. Quadrupeds, such as this woolly spider monkey, usually have limbs of about the same length.



CLINGING
Vertical clingers and leapers, such as these indri, move with the back held vertically. They have well-developed back limbs for long leaps between trees.



BALANCING ACT
The chimpanzee stabilizes the termite-covered stick by resting it on the opposite forearm.

EATING QUICKLY
Most of the termites, including any on the forearm, are swiftly gobbled before they can escape.

MORE OF THE SAME
Once all the termites have either been eaten or escaped, the chimpanzee will repeat the process.

LEARNED BEHAVIOR
Making and using tools is rare in mammals. Juvenile chimpanzees acquire these skills by copying adults.